# Order and Signalling in Communicating Quantum Programs: a Semantics of Causality

Qualifying Dissertation for the Transfer of Status

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Michaelmas Term 2026

**Abstract**

A quantum program that sends and receives qubits has a causal structure: the order of its communication events, and the question of which of them can influence which. This thesis develops a programming-language semantics in which that structure is denotational data and is characterised extensionally by the signalling types of higher-order quantum theory. It builds on an algebraic theory of classically controlled quantum communication, begun in my MSc dissertation, whose operational and denotational models are adequate and fully abstract for each other and for which I have a draft proof of completeness; the proposed work gives this semantics its signalling reading as a family of combs, extends it to several channels, and tests a hypothesis that would make the finished semantics of causal probabilistic programming from my first year, in which interventional causal structure is characterised by contextual equivalence, literally a fragment of it: that an intervention is a communication event with the intervener as the environment. Part I reviews the literature the thesis builds on, in programming-language semantics and causal inference, and positions the two neighbouring fields it reaches into, the semantics of concurrency and higher-order quantum processes. Part II states the research plan and is self-contained.

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# Part I
# Literature Review

## 1 Introduction

**Three notions of causality.** When a computer scientist says that one part of a system *causes* another, they may mean at least three distinct things. In Pearl's theory of causal inference [105], $X$ has a causal effect on $Y$ if *intervening* on $X$—forcing it to a value, regardless of how it would otherwise have been produced—changes the distribution of $Y$. In quantum foundations and in the theory of higher-order quantum processes [80, 100], the causal structure of a process is given by its *signalling* constraints, which inputs can influence which outputs, together with the further question of whether it is compatible with a definite causal order between its parts. In probabilistic modelling, a causal structure is a directed acyclic graph along which a joint distribution *factorises*, and the content of the structure is the family of conditional independences it entails [105, 53].

Three further usages should be named so as to be set aside. In categorical quantum mechanics, a process is called *causal* when it preserves the discarding map [40]. This is the causality axiom of Chiribella, D'Ariano and Perinotti [33], glossed as "no signalling from the future", and Coecke [39] shows that it implies the no-signalling principle for processes composed along a causal structure; it is the base axiom on which signalling structure is built, not a notion of causality between parts. Fritz [52] calls a Markov category *causal* when the output of a morphism cannot depend on a choice made downstream of it; this is again a property of the ambient category. In artificial intelligence and in verification, *actual causality* [63] asks which particular event caused which particular outcome in a single run; that question is not treated here. Nor are experimental tests of indefinite causal order, causal discovery, or identifiability, each of which is mentioned below only where a work package touches it.

That the same word carries these meanings is not an accident, but neither is it a theorem. Each notion is *extensional*: it characterises causal structure by what an observer may do to the system—intervene, send signals, condition. What such observers probe is *intensional*: the dataflow

of a program, the temporal order of its communication events, the graph of a model. Programs carry this intensional structure explicitly in their syntax. The viewpoint of this thesis is therefore that causality is a property of *program semantics*, and that the right way to relate the three notions is to relate, for each one, the intensional structure in the syntax to the extensional characterisation in the semantics by a correspondence theorem.

**Where the two routes stand.** Two lines of work precede this proposal, and the review is organised around them. On the interventional side, my coauthors and I built a hierarchy of first-order probabilistic languages in which Pearl's interventions and counterfactual *twinning* are program transformations, defined program equivalences contextually, and gave a graded denotational semantics that is fully abstract for concrete models (§3). This route has the complete picture for one notion of causality: the intensional dataflow of a program is characterised extensionally by equivalence under all interventional contexts. On the communication side, my MSc dissertation gave an algebraic theory $\mathsf{IO} \otimes \mathsf{Q}$ of classically controlled quantum communication, an operational model in which programs exchange qubits with a quantum stream, and a denotational model given by a free I/O monad on $\mathbf{CP}^{\infty}$, with the denotational model adequate and fully abstract for the operational one; this year's work, with my supervisor, is its conference write-up and a draft proof of completeness (§4). Here the temporal order of communication events is first-class semantic data: the denotation is indexed by I/O traces. This route does *not* yet have an extensional causal reading of that order; giving it one, as signalling structure in the sense of higher-order quantum processes, is part of the research plan rather than a result.

A single example shows why these are two faces of one question. In the interventional language, a program reads an *intervention label* $\ell$ through $\mathsf{applyIntv}\ \ell\ t$, which yields the label's value when it is set and $t$ otherwise; if a second variable $z$ does not depend on the result, every intervention at $\ell$ leaves the distribution of $z$ unchanged, and the contextual equivalence makes this an equation between programs. Causal independence is *tested by contexts*. In the theory $\mathsf{IO}\otimes\mathsf{Q}$, the commutation axiom $\mathsf{apply}_U(\boldsymbol{a},\ \boldsymbol{a}.\ \mathsf{in}(b.\ x(\boldsymbol{a},b))) = \mathsf{in}(b.\ \mathsf{apply}_U(\boldsymbol{a},\ \boldsymbol{a}.\ x(\boldsymbol{a},b)))$ says that a local operation and a communication event on different qubits may be reordered, and the quantum-stream semantics validates it; by contrast $\mathsf{in}$ and $\mathsf{out}$ do not commute with one another, and that is exactly why the trace index is a total order. Causal independence is *an equation validated by an observer*. Both statements say that no information flows from one place to another, so that reordering or intervening is unobservable; in programming-language terms both are *noninterference* properties [60]. Whether this coincidence is an instance of a general correspondence is the question of Part II.

**Scope of the review.** Sections 2–4.3 treat the literature on which the thesis builds: the semantics of effects, probabilistic programs and quantum programs; causal models, their equivalences and their diagrammatic presentations; causal probabilistic programming; and quantum process calculi. Sections 4.4 and 4.5 position two neighbouring fields that the research plan reaches into, the semantics of concurrency and higher-order quantum processes, at the level of their main definitions and results; their detailed survey is a milestone of that plan (Part II, §12). Section 5 draws the three notions together and identifies the gap the thesis addresses.

# 2 Semantic foundations shared by both routes

## 2.1 Monads, algebraic theories, and their combinations

Moggi's observation that notions of computation are monads [95, 96] and Plotkin and Power's refinement that most of them are presented by algebraic theories of operations and equations [109, 110] are the common foundation of both routes. An algebraic theory determines a monad, and conversely a finitary monad is a Lawvere theory [69]; the equational presentation is what makes program equivalence a matter of derivability. Hyland, Plotkin and Power [68] showed that the standard ways of combining theories have computational meaning: the *sum* adjoins the operations of two theories with no interaction, so that the order of effects from the two theories is observable,

while the *tensor* additionally imposes that operations of one theory commute with those of the other, so that their relative order is unobservable. In the premonoidal semantics of effects [112, 113, 55], the morphisms that commute with everything are the *central* ones; centrality is the established algebraic form of "cannot be observed to happen before or after". Handlers [14] give the programming counterpart of models; comodels and runners [111, 6] describe the environments in which effectful programs execute.

Ordinary algebraic theories cannot express operations that create and consume *resources*, such as qubits, which by no-cloning must be treated linearly. Staton's *parameterised algebraic theories* (PATs) [134, 135] extend the framework with parameters: operations consume parameters, classically choose a branch, and bind fresh parameters in each branch. Staton used them to give a fully complete equational theory of qubit quantum computation [135]; Stark's free model of $\pi$-calculus name creation [133] is the earlier instance of the same commutation discipline, where communication primitives commute with independent local operations. Kammar, Liell-Cock, Lindley, Matache and Staton [76] have given a complete equational axiomatisation of dynamic threads (fork, wait, stop, atomic actions) as a parameterised algebraic theory, whose closed terms correspond exactly to pomsets; I return to this in §4.4, since it is the nearest neighbour of my own completeness result.

*Where we stand.* My MSc dissertation [141] shows that sums and tensors exist for the enriched Lawvere theories presented by PATs, by explicit syntactic construction, which is what licenses the definition of $\mathsf{IO} \otimes \mathsf{Q}$ as a tensor in §4.

*What this gives us.* The tensor, and centrality more generally, are already primitive statements about causal independence. In $\mathsf{IO} \otimes \mathsf{Q}$ the tensor makes communication commute with local operations on other qubits, while $\mathsf{in}$ and $\mathsf{out}$ do not commute with each other. I treat this as an observation to be made precise, not as a result: Part II asks (WP 3) whether centrality in an algebraic theory is in general the algebraic shadow of no-signalling.

## 2.2 Semantics of probabilistic programs

Probabilistic programming gives a compositional syntax for statistical models and performs Bayesian inference on them compositionally [91]. Its denotational semantics is monadic: the Giry monad on measurable spaces [81, 136], and, for higher-order programs, the probability monad on quasi-Borel spaces [64]. Implementations such as LazyPPL [42], MonadBayes [123, 124] and GenJAX [15] realise this monadic structure directly. Full abstraction is the recurring theorem shape for such languages, the standard reference point being probabilistic coherence spaces for probabilistic PCF [49].

The synthetic approach of *Markov categories* [52, 54] axiomatises the categorical structure needed to talk about probability without measure theory: a symmetric monoidal category with a commutative comonoid on every object, where discarding is natural (the unit is terminal) but copying need not be. In Markov categories with conditionals, disintegration and Bayesian inversion are expressed synthetically [36, 52], and Fritz and Klingler [53] prove the soundness of d-separation, and the equivalence of the local and global Markov properties, synthetically; completeness of d-separation in the sense of Geiger, Verma and Pearl remains the classical result. This is the *factorisation* face of causality: a Bayesian network is a string diagram in a Markov category, and the conditional independences it entails are read off the diagram.

The second piece of structure both routes rely on is *grading*. Graded monads [78, 98, 56, 46] and, more generally, locally graded categories [87, 121] track an index—an effect annotation, a trace, a resource—through composition. Milius, Pattinson and Schröder [92] use graded monads for trace semantics; Liell-Cock and Staton's **ImP** construction [88] uses grading to make imprecise probability compositional. The interventional semantics of §3.5 is built on an analogous construction in which the grade records which interventions are available.

*What this gives us and what it lacks.* Markov categories give factorisation but no primitive notion of intervention; the diagrammatic causal-model literature (§3.3) adds it by surgery on diagrams. Neither offers a notion of program *context*, which is what contextual equivalence needs.

### 2.3 Semantics of quantum programs

Selinger's QPL [125] established the denotational template for quantum programming languages: types denote finite-dimensional spaces of density matrices (or, more generally, $C^*$-algebras [35]), and programs denote completely positive maps. Categorical quantum mechanics [3, 65, 40] gave this a graphical calculus through dagger compact closed categories and the CPM construction [126], and the ZX-family of calculi axiomatise fragments of it; Staton's PAT Q [135] and Carette et al.'s extension of Π [27] give complete equational theories of qubit programs and of unitary circuits respectively. Languages such as QWIRE, Quipper and Silq [104, 61, 19] work in the circuit model with classical control. For languages with infinitary features one needs the completion of **CP** under infinite biproducts, $\mathbf{CP}^\infty$ [102, 150], which is the base category of the communication semantics. Quantum monads [122, 8] and quantum game semantics [37] are two further lines of work on the structure of quantum computation as an effect; the latter uses event structures and is the closest existing semantics to the one proposed in WP 2a.

*What this gives us.* The axiom that singles out the deterministic (discard-preserving) processes in categorical quantum mechanics [40, 33] is the base condition from which the signalling structure of §4.5 is built; the **Caus** construction requires compact closure and further "precausality" axioms on top of it, and, by design, its base category also contains the non-deterministic, trace-decreasing processes.

## 3 Causality as intervention

### 3.1 Structural causal models and Pearl's hierarchy

Pearl's structural causal models (SCMs) [105, 106] are systems of equations $X_i = f_i(\mathrm{Pa}_i, U_i)$ in which each endogenous variable is a deterministic function of its parents and of an exogenous noise variable; a prior on the noise makes an SCM a probabilistic model, and the equations make it more than a distribution. The surplus is organised by the *causal hierarchy* [11]: rung 1 asks observational questions; rung 2 asks *interventional* questions, answered by replacing the equation for $X$ with a constant (the do operator); rung 3 asks *counterfactual* questions about a particular run, answered by abduction, action and prediction, implemented graphically by *twinning*, which duplicates the model into worlds that share the noise [10]. Both operations are *syntactic transformations of the model*. The Neyman–Rubin potential-outcomes framework [120] is the other dominant formalism; its correspondence with SCMs is well understood [72]. Richardson and Robins' single-world intervention graphs [117] present an intervention by splitting a node into its natural and its fixed value, which is the graphical counterpart of the intervention labels of our languages; they are deliberately single-world, and cross-world quantities such as natural direct effects need further assumptions there. Interventions need not be perfect: the literature on stochastic and soft interventions [43, 44, 48, 86] replaces an equation by another mechanism rather than a constant.

Throughout, I work with fully specified models and Bayesian inference: mechanisms and priors are given, and queries are answered by inference. *Identifiability* (when an interventional quantity is computable from observational data, the subject of the do-calculus [128, 127]) and *causal discovery* [132, 108] are outside the scope of this thesis.

*What this gives us.* The causal structure of an SCM is its dataflow: which variables each equation reads. Interventions probe it. The semantic question is when two models have the same causal structure, and the answer must be extensional.

### 3.2 Equivalences of causal models and logical accounts

Bongers, Forré, Peters and Mooij [23] give the principal account of equivalence for SCMs at all three rungs: two models are observationally, interventionally or counterfactually equivalent when they agree on the corresponding class of queries, with the counterfactual case defined through the two-world twinned model. Their treatment also covers cyclic SCMs, through solution functions. At rung 2, several further notions exist: Pearl's causal Bayesian networks are families of interventional

distributions compatible with a graph; *causal spaces* [103, 25] axiomatise this measure-theoretically; and *causal abstraction* [119, 16, 118, 90, 50] studies coarsening maps between models that commute with interventions. At the level of graphs rather than models, Markov equivalence [138] is what discovery algorithms can recover. On the logical side, Ibeling and Icard [71] characterise the three rungs by three probabilistic logics of strictly increasing expressivity whose semantics is defined on SCMs through Pearl's operations, and Icard has proposed probabilistic Turing machines as the computational counterpart of SCMs [73, 70].

*What this gives us and what it lacks.* These are all extensional characterisations of interventional structure, and they are the yardstick against which our contextual equivalences are measured (§3.5). None of them is formulated as a contextual equivalence of programs, and none treats higher-order models.

### 3.3 Diagrammatic and categorical causal models

Fong's *causal theories* [51] present a Bayesian network as a symmetric monoidal category freely generated by its graph, with models as functors into stochastic maps. Jacobs, Kissinger and Zanasi [74] present interventions as *string-diagram surgery* in this setting, and introduce *comb disintegration*: the use of combs, processes with a hole, to express the classical causal-inference computations, which is the first appearance of the comb notion of §4.5 in classical causal inference. Lorenz and Tull [89] develop causal models in string diagrams systematically, and a later line presents causal abstraction as natural transformations between such models [90, 50]. Bayesian inference in this style is handled by *normalisation boxes*, which renormalise locally inside a diagram [75], and Agrawal et al. [5] obtain equational principles for causal Bayesian networks from them. Zhang et al. [149] characterise d-separation as a noninterference-style separation property, which makes explicit a connection implicit in all of this work and in §1: invariance under interventions is a noninterference property [60].

*What this gives us and what it lacks.* This is the closest existing bridge between the factorisation face and the intervention face of causality, and it is compositional in the functorial sense. Its equational theories are, however, stated per graph, and it has no notion of program context or of higher-order models. It is where the unifying work package (WP 3) starts, and comb disintegration is the prior work that WP 3a must be positioned against.

### 3.4 Causal probabilistic programming languages

Causal probabilistic programming aims to make intervention and twinning primitives of an expressive probabilistic language. Libraries such as DoWhy [116, 22] and the `do` operators of Pyro and PyMC [20, 4] manipulate models as graphs or computation graphs and stop at rung 2 (DoWhy's GCM module reaches rung 3 on fitted mechanisms). Witty et al. [146, 147] treat an intervention as a source transformation that replaces subterms, which is powerful but non-compositional; Brulé's Whittemore [24] is an early embedded causal DSL.

Four systems reach rung 3 natively. OmegaC [137] obtains interventions by a lazy evaluation strategy that retroactively modifies closures, reminiscent of dynamic binding, and counterfactuals by making every program a deterministic function of explicit noise. ChiRho [30] implements both as effect handlers over Pyro, with a multi-world value type whose relation to Pearl's twinning is not immediate. MultiVerse [107] uses trace inspection and re-execution. Kiesel et al. [79] answer counterfactual queries on ProbLog programs by a transformation proved correct against Pearl's semantics.

*What this gives us and what it lacks.* These systems show that the primitives are implementable at higher order, by non-standard evaluation techniques. What they lack is a denotational and equational foundation: a semantics in which intervention is a mathematical operation rather than an evaluation strategy, and an account of when two causal programs are equivalent. That is what the first route supplies.

### 3.5 Where we stand on the interventional route

Our POPL submission [FILL: status: under revision for POPL 2027?; coauthors] (an early version appeared at the LAFI workshop [140]; artifact [139]) reconstructs Pearl's hierarchy as a hierarchy of three first-order probabilistic languages. $L_1$ is a standard first-order probabilistic language with conditioning by a final score. $L_2$ exposes hooks in the dataflow as typed free variables, the *intervention labels*, read through $\mathsf{applyIntv}$; an intervention $\mathsf{do}(\ell \mapsto v)\{t\}$ is substitution of $v$ for $\ell$. $L_3$ annotates each binding as exogenous or endogenous, and *twinning* $\mathsf{twin}_n$ is a vectorisation transformation from $L_3$ to $L_2$ that duplicates the endogenous bindings into $n$ worlds sharing the exogenous ones. Program equivalence at rung $k$ is *contextual* with respect to rung-1 denotational equality, in Milner's sense: two programs are $\equiv_k$ when they are denotationally equal in every rung-$k$ context, and varying the admissible interventions and world counts gives a spectrum of equivalences. For representable acyclic SCMs compiled into $L_3$, the observational, interventional and counterfactual equivalences of Bongers et al. coincide with the equivalences $\equiv_3^{\text{obs}}$, $\equiv_3^{\text{intv}}$ and $\equiv_3^{\text{2-world}}$ of the compiled programs; the full $\equiv_3$ quantifies over all $n$-world twinnings, and whether it is strictly finer than $\equiv_3^{\text{2-world}}$ is open. The denotational semantics interprets interventions by *grading*, in a construction reminiscent of **ImP** [88], and twinning by world-polymorphic vectorisation. It is adequate for the contextual equivalences unconditionally and fully abstract when the underlying model is *concrete*, meaning that values correspond to global elements and global elements separate morphisms; **Set**, standard Borel spaces and quasi-Borel spaces are concrete, and the construction is parametric in the model. The denotation then motivates a higher-order language $L_{\mathsf{HO}}$ with cross-world interventions, into which $L_3$ compiles in a semantics-preserving way, and which has a definitional Haskell implementation. [FILL: one sentence on which parts are yours: e.g. the contextual-equivalence framework and the full-abstraction proofs]

The limitations relevant to Part II are these: the languages are acyclic; the relation between the contextual equational theory and the normalisation-box equational theories of §3.3 is not worked out; the first-order semantics is over normalised probability, while $L_{\mathsf{HO}}$ widens to the unnormalised measure monad to internalise conditioning, so that programs with observations denote sub-normalised maps; and the models are measure-theoretic rather than finite-dimensional, which matters for the signalling face (§5).

## 4 Causality as order and signalling in communicating quantum programs

### 4.1 Quantum communication and distributed quantum computing

Quantum communication underlies teleportation [18], key distribution [17] and, increasingly, proposals for distributed quantum computing that connect small quantum processors into larger ones [26]. Programming models for distributed quantum computing are at present circuit-level: they describe which gates are applied where and which qubits are sent when. On the semantic side, Ceragioli, Gadducci, Lomurno and Tedeschi observe that the existing quantum process calculi do not agree on what the observational properties of quantum values should be [29]. There is, in other words, no agreed account of what a distributed quantum program *means*, nor of which of its behaviours are physically realisable.

### 4.2 Quantum process calculi

Quantum communication has mostly been studied through quantum process calculi—CQP [57], QPAlg [83], qCCS [148] and lqCCS [29]—which extend classical calculi [93, 66, 94] with qubit message passing. They are presented operationally, by labelled transition systems over configurations, with program equivalence given by bisimilarity. As the authors of lqCCS show, the proposed bisimilarities differ in subtle ways because of the interaction of concurrency, synchronisation and local nondeterminism with quantum effects; their later coalgebraic account of effect-labelled transition

systems [28] is an attempt to organise these subtleties. A separate difficulty is well known from the classical theory: parallel composition is not an algebraic operation and cannot be adjoined to an algebraic theory as an operation [133, 59], although workarounds exist, such as the cooperative threads of Abadi and Plotkin [1].

*What this gives us and what it lacks.* Process calculi are the only existing semantic treatment of communicating quantum programs, and they are expressive. They are operational only, their equivalences are not canonical, and they entangle communication with concurrency, which makes it hard to isolate what is specific to *communication*. My work studies communication in isolation, as an algebraic effect, and is in that sense the local sublanguage of these calculi with a denotational semantics.

### 4.3 Where we stand on the communication route

My MSc dissertation [141], and its conference version with my supervisor [FILL: FoSSaCS submission status], introduce $\mathsf{IO} \otimes \mathsf{Q}$, an algebraic theory of classically controlled qubit communication, as the *tensor* of a standard I/O theory [68] with Staton's complete theory $\mathsf{Q}$ [135]: programs may send and receive qubits depending on measurement outcomes, and communication commutes with local quantum operations. The tensor is argued to be needed, since one equation that holds in the tensor but not in the sum is required to validate a quantum phenomenon involving entanglement across the program–environment boundary (Example B in Part II). The theory is given two models. The operational model lets a program exchange qubits with an environment modelled as a *quantum stream*; a program equivalence $\approx_q$ compatible with quantum theory is defined from it by inspecting final quantum states, and terms modulo $\approx_q$ form a model of $\mathsf{IO} \otimes \mathsf{Q}$, which shows that the tensor does not collapse in the manner of Eckmann–Hilton. Although operational, $\approx_q$ is therefore not a contextual equivalence; whether a contextual equivalence observing only traces, continuations and a final measurement coincides with it is a question I return to in Part II. The denotational model is the free I/O monad $T(X) = \bigoplus_\sigma \mathrm{qbit}^{|\sigma|} \otimes X$ on $\mathbf{CP}^\infty$, indexed by I/O traces $\sigma$. The main theorem is that the denotational semantics is adequate and fully abstract for the operational one: $[\![c]\!] = [\![c']\!]$ iff $c \approx_q c'$. What the dissertation left as a conjecture is *completeness*: that $[\![t]\!] = [\![t']\!]$ implies $t = t'$ is derivable in $\mathsf{IO} \otimes \mathsf{Q}$. I now have a draft proof of this (manuscript in preparation, not yet closed), by induction on the I/O depth of terms, using a reorganisation step that normalises every term into an I/O-free head applied to a single input and a single output branch, and a factorisation lemma on completely positive maps that lets semantically equal terms be rewritten to a common prefix. If it closes, the three semantics—algebraic, operational, denotational—coincide.

The observation that sets up Part II is this. The trace $\sigma$ is a *temporal* order of communication events, exposed in the denotation: the $\sigma$-component of a closed term is a completely positive map with $|\sigma|$ open qubit wires, one per event, to be used in the order $\sigma$, and when the term branches on measurements this component is sub-normalised, its trace being the probability of the branches that produce $\sigma$. Whether the family of these components is a *causal* object in the signalling sense of §4.5, a comb or a tester whose slots are ordered by $\sigma$, is a question the dissertation raised and did not answer. It is stated as Conjecture 1 in Part II.

### 4.4 Order in concurrency theory

Concurrency theory has long modelled causal order explicitly. Lamport's happens-before relation [84] is the partial order an observer of a distributed system can reconstruct. Mazurkiewicz traces [45] are words modulo the commutation of independent letters, the cheapest partial-order model and the one that arises when independent channels are added to a sequential program. Pratt's pomsets [114]—partially ordered multisets of events, up to isomorphism, with series and parallel composition—generalise words to partially ordered words, and Gischer [58] axiomatised the equational theory of series-parallel pomsets. Winskel's event structures [144, 97, 145] add a conflict relation to the causal order and give a domain-theoretic account of nondeterministic concurrent computation. Concurrent Kleene algebra [67] adds a commutative parallel operator to Kleene al-

gebra, related to sequential composition by an exchange law; Kappé, Brunet, Silva and Zanasi [77] prove that series-parallel pomset languages closed under Gischer's subsumption order are its free model, by reduction to an earlier completeness theorem for bi-Kleene algebra [85]. Within the algebraic-effects tradition, Dvir, Kammar and Lahav [47] give a monadic semantics of shared-state concurrency from an equational theory, and Kammar et al. [76] axiomatise dynamic threads as a parameterised algebraic theory whose closed terms are exactly pomsets, with a completeness theorem. Clairambault, de Visme and Winskel [37] give an event-structure game semantics for quantum programs.

*Position.* The trace-indexed monad of §4.3 is the degenerate single-channel case of this picture: a single channel forces a total order, so traces are words. Two extensions are distinguished in Part II. Adding independent channels to a *sequential* program makes events on different channels commute, so that traces become Mazurkiewicz traces and the monad a free monad over a partially commutative monoid; this keeps the proof architecture of the single-channel case. Adding *threads* makes traces pomsets, and the theory of Kammar et al. is a parameterised algebraic theory with exactly that free model, so "tensor $\mathsf{IO} \otimes \mathsf{Q}$ with a thread theory" is a well-posed, and much harder, second step. Completeness results in this literature also go through free models; whether their techniques transfer to the quantum setting, where a factorisation lemma on CP maps did the work, is part of the first reading milestone of the plan. The logics of session types are the other point of contact: BV and its relatives have been used as logics of multiparty protocols, and Simmons conjectures that causal consistency of a wiring of black boxes corresponds to deadlock freedom of the protocol [129]; a trace-indexed semantics of communicating programs is where such a correspondence could be made precise.

### 4.5 Higher-order quantum processes and signalling

Higher-order quantum computation studies transformations of quantum processes rather than of states. Chiribella, D'Ariano and Perinotti's *quantum combs* [31, 32] are processes with holes in a fixed causal order; their probabilistic counterparts, *testers*, and Gutoski and Watrous's *strategies* [62] describe multi-round interactions with sub-normalised outcomes, and the general theory of supermaps forms a hierarchy of types [21]. Oreshkov, Costa and Brukner's *process matrices* [100] are the most general bipartite process operators compatible with quantum mechanics holding locally in each party's laboratory; the multipartite theory and the notion of *causal separability*, in which the order of later parties may depend classically on earlier outcomes, are due to Araújo et al. and Oreshkov and Giarmatzi [9, 101]. Some process matrices are causally non-separable. Two kinds of evidence distinguish them: device-independent violation of a *causal inequality*, which Oreshkov, Costa and Brukner's process exhibits, and device-dependent witnesses of non-separability. The *quantum switch* [34], which applies two operations in an order controlled by a qubit in superposition, is causally non-separable [9] but produces only causal correlations; photonic experiments implement it [115], and whether such implementations realise or simulate indefinite order is debated [99, 142].

Which indefinite-order processes can be built as circuits is the subject of Wechs, Dourdent, Abbott and Branciard [143], who define nested classes of quantum circuits with fixed order, with *classical* control of causal order (the next operation may depend on classical outcomes of earlier ones) and with *quantum* control of causal order, and characterise each by conditions on process matrices. The switch lies in the quantum-control class and not in the classical-control one; every classical-control process is causally separable; and quantum-control processes generate only causal correlations. That the quantum-control class exhausts the physically realisable processes is a proposal, not a theorem.

Categorically, Kissinger and Uijlen's **Caus** construction [80] builds, over a *precausal* category—compact closed, with a chosen discarding structure and further axioms, the examples being **CP** on finite-dimensional systems and $\mathbf{Mat}(\mathbb{R}_{\geq 0})$, finite matrices of non-negative reals—a $*$-autonomous category whose types are signalling constraints: first-order processes, non-signalling processes, one-way signalling processes, and so on. The dual of the non-signalling type contains all bipartite process matrices, causally separable or not; separability is a property inside that type, not a type, and

Simmons proves that every process matrix is an affine combination of definite-order processes [129]. Simmons's thesis [129] is the fullest development of this line and the reference for most of what Part II needs on the signalling side. It rebuilds the construction over *additive* precausal categories (compact closed with products, hence biproducts, and a finite causal basis for every object), which gives **Caus** products, coproducts and probabilistic choice as type operators; it adds a non-commutative sequential operator for one-way signalling, under which **Caus** models Guglielmi's BV-logic [130]; it defines $n$-ary *graph types* encoding compatibility with an arbitrary DAG of signalling relations; it surveys candidate ways of admitting sub-normalised, partial processes into the construction, effectus-style, and where each falls short; and it introduces *causal logic*, pomset logic with directed axiom links, whose proof nets characterise exactly the causally consistent wirings of black boxes [129, 131], with BV and pomset logic separated by a process-matrix example. The finite-causal-basis axiom excludes infinite-dimensional and measure-theoretic bases, which is one of the two obstructions discussed in §5. Finally, *quantum causal models* transfer Pearl's framework to quantum processes: Costa and Shrapnel [41] gave the first process-matrix-based definition, Allen et al. [7] the common-cause line, and Barrett, Lorenz and Oreshkov [12] the general framework, in which a node is an input–output pair, the process operator factorises as a product of commuting operators (a quantum Markov condition), interventions are local instruments at nodes, and d-separation in the graph implies conditional independence of classical outcomes. The cyclic extension [13] gives a cyclic-causal-model account of some processes without definite order, including the switch. Coherent control of channels [2, 82] and, on the language side, the PBS-calculus of Clément and Perdrix [38], which has the switch as a term, are the relevant prior work on coherent control.

*Position.* Three things follow for this thesis. First, the family of $\sigma$-components of an $\mathsf{IO} \otimes \mathsf{Q}$ denotation has the shape of a tester or strategy whose slots are ordered by the traces; that it *is* one is Conjecture 1. Second, signalling types are the extensional face of order in the same way that interventional equivalence is the extensional face of dataflow, and quantum causal models are the one existing place where factorisation, intervention and signalling are all present, which is why they are the first reading milestone. Third, the analogy between parallel composition and indefinite causal order must be stated with care, and the process-matrix picture is the right way to state it: the parties' local programs are combs, and the process matrix is the *network* that wires them together, in other words the scheduler. Parallel composition $P \parallel Q$ leaves the order of events unspecified. A classical scheduler, even one whose later choices depend on earlier outcomes, yields a process in the classical-control class, hence a causally separable one. The switch is a coherent scheduler, well-typed on every non-signalling pair of channels, whose output applies the two operations in a superposition of orders; it is causally non-separable, and it has content only when the two programs act on a shared target or communicate, since on disjoint systems every order gives the same channel. The analogy is therefore between $\parallel$ and the argument type of a scheduler, not between $\parallel$ and the switch itself; this is what makes WP 2d well-posed, and also what makes it hard, since coherent control of order is not an algebraic operation with classical continuations. Finally, Simmons's own outlook [129] proposes reading send and receive operations as physical processes so as to assign causal types to communicating programs, with causal consistency playing the role of deadlock freedom and the session-type logics BV, GV and MAV as the point of contact; that is, from the physics side, the direction of WP 2a and WP 2b.

## 5 Synthesis: three notions of causality and the gap

Table 1 summarises the review. Each column is one extensional notion of causality; each row asks the same question of it.

Three remarks. First, the example of §1 and the asymmetry it exposes: for intervention the correspondence between intensional and extensional structure is a theorem; for signalling there is a semantics with order as first-class data and a conjecture. Second, non-DAG structure is the next hard case in every column, but it is not the same case: feedback in cyclic SCMs, causal non-separability in quantum processes, and unscheduled parallel composition in programs are three

| | **Intervention** | **Signalling** | **Factorisation** |
|---|---|---|---|
| Intensional datum | dataflow of a program / SCM | temporal order of events: trace $\sigma$, comb order, pomset | DAG |
| Extensional characterisation | contextual equivalence under interventions | signalling types (**Caus**); causal (non-)separability | conditional independence / d-separation |
| Correspondence theorem | full abstraction for concrete models (§3.5) | full abstraction and draft completeness relate three equalities (§4.3); the signalling reading is Conjecture 1 | soundness of d-separation, synthetically [53]; completeness classically |
| Semantics of programs | first-order and higher-order, acyclic | single channel, classical control | Markov categories; interventions by surgery |
| Next hard case | cyclic SCMs | multiple channels; causally non-separable processes | cyclic and quantum causal models |

Table 1: Three notions of causality and where the two routes stand.

different departures from a fixed order, and a unifying account must distinguish them. Third, and this is the gap: there is no framework in which the three columns are structure on one category. Markov categories cover factorisation and, through surgery, part of intervention; **Caus** covers signalling for quantum and finite classical processes; quantum causal models relate all three, with a compositional presentation in Lorenz and Tull's string diagrams, but are not formulated as a semantics of programs. The two semantics of §3.5 and §4.3 cover the first two columns separately, and both are families of morphisms indexed by an intensional datum, intervention grades or traces, with a reindexing action, which is the common structure a generalisation would start from.

Two obstructions should be displayed rather than hidden. The first is *finite dimension*: the **Caus** construction needs a compact closed base with invertible dimensions, which $\mathbf{Mat}(\mathbb{R}_{\geq 0})$ and finite-dimensional **CP** provide, but measurable spaces, quasi-Borel spaces and the infinite-biproduct completion $\mathbf{CP}^{\infty}$ do not. The bridge of WP 3a is therefore stated on the finite fragment, and the comb reading of WP 2b is stated per trace, over finite-dimensional **CP**, with the trace index kept external. The second is *sub-normalisation*: conditioning in $L_{\mathsf{HO}}$ denotes sub-normalised maps, and branching in $\mathsf{IO} \otimes \mathsf{Q}$ makes individual trace components sub-normalised, so neither lands in a first-order **Caus** type as it stands; testers and strategies [32, 62] are the objects that accommodate sub-normalised families, Simmons [129] surveys the ways of admitting partial processes into **Caus** and finds each wanting in some respect, and normalisation boxes [75] are the diagrammatic tool for renormalising after conditioning. I also note that the factorisation column is the thinnest for me: it enters through Markov categories and the diagrammatic causal-model literature, not through my own results, and WP 3a includes a factorisation reading for that reason.

# Part II
# Thesis Proposal

## Notation and background in brief

- *Communication theory* $\mathsf{IO} \otimes \mathsf{Q}$: the tensor of an I/O theory ($\mathsf{in}$ receives a qubit, $\mathsf{out}$ sends one) with Staton's theory $\mathsf{Q}$ ($\mathsf{new}$, $\mathsf{measure}$ with two branches, $\mathsf{apply}_U$). Binding: $\mathsf{apply}_U(\boldsymbol{a},\ \boldsymbol{a}.\ t)$ applies $U$ to $\boldsymbol{a}$ and continues as $t$; $\mathsf{discard}(q, t)$ measures and ignores. $\approx_q$: operational equivalence against quantum streams. $T(X) = \bigoplus_{\sigma} \mathrm{qbit}^{|\sigma|} \otimes X$ on $\mathbf{CP}^{\infty}$: the free I/O monad; $[\![t]\!]_{\sigma}$: the $\sigma$-component of a denotation.
- *Signalling*: a *comb* is a process with slots in a fixed order; a *tester* is a sub-normalised family

of combs summing to one. **Caus**($\mathcal{C}$): Kissinger and Uijlen's category of higher-order processes over an additive precausal $\mathcal{C}$, typed by signalling constraints; **Mat**($\mathbb{R}_{\geq 0}$) and finite-dimensional **CP** qualify; graph types [129] are the signalling types of an arbitrary DAG; causal logic [129, 131] is the logic whose proof nets are the causally consistent wirings.

- *Interventional languages* $L_1 \subseteq L_2 \subseteq L_3$ and $L_{\mathsf{HO}}$: first-order probabilistic languages with intervention labels $\ell$ read by $\mathsf{applyIntv}$, interventions $\mathsf{do}(\ell \mapsto v)\{t\}$ by substitution, twinning by vectorisation, and a higher-order extension. $\equiv_k$: contextual equivalence at rung $k$. **FinStoch**: finite stochastic matrices, the first-order part of **Caus**(**Mat**($\mathbb{R}_{\geq 0}$)).
- M1–M2: reading milestones in §12.

# 6 Research topic and fundamental challenges

The thesis will develop a *semantics of order and signalling for communicating quantum programs*: a programming-language account of what a communicating quantum program means, in which the temporal order of its communication events is denotational data, and that order is characterised extensionally by the signalling types of higher-order quantum theory. The finished interventional work of §3.5 is the template for this account and its comparison target: a semantics in which causal structure is characterised by a contextual equivalence, proved fully abstract. The thesis's central hypothesis is that the two are instances of one thing, that *an intervention is a communication event with the intervener as the environment*, and the thesis closes by testing that hypothesis on the finite classical fragment where both semantics are defined.

The fundamental challenges are these. (i) *Sub-normalisation.* Branching on measurement outcomes makes individual trace components sub-normalised, so they are not combs, and the signalling reading must be stated for the family of components, as a tester, rather than componentwise. (ii) *Finite dimension.* Signalling types are built over compact closed categories with a finite causal basis, which the infinite-biproduct completion $\mathbf{CP}^{\infty}$ does not have, so the comb reading must keep the trace index external. (iii) *Parallel composition is not an algebraic operation* [133, 59]: independent channels can be added to a sequential theory, threads cannot, and coherent control of order is not an operation with classical continuations at all. (iv) *Completeness proofs are long and model-specific*: the single-channel proof has taken most of a year and is not yet closed. (v) *No accepted standard to test against*: the existing behavioural semantics for communicating quantum programs disagree with one another [29].

# 7 Goal

I want a mathematical account of what a quantum program that sends and receives qubits means, in which "this event can influence that one" is something the semantics states and checks, and which turns out to be the same account as the one that already works for "if I change this, that changes" in probabilistic programs.

# 8 State of the art and why the problem is hard

Quantum process calculi (CQP, qCCS, lqCCS) are the only semantic treatment of communicating quantum programs; they are operational, their bisimilarities disagree, and they do not separate communication from concurrency. The theory of higher-order quantum processes (combs, process matrices, the **Caus** construction and its graph types and causal logic) has a precise account of signalling and of definite versus indefinite order, but it has never been connected to a programming language; its outlook proposes exactly that connection, through send and receive operations read as physical processes [129]. On the interventional side, notions of equivalence for causal models (Bongers et al.; causal spaces; causal abstraction) are not equivalences of programs, and causal probabilistic programming languages implement interventions by evaluation strategies without an

equational account; my own prior work closes that gap and is the template here. Markov categories, the synthetic home of factorisation, have no primitive notion of intervention or of signalling.

The problem is hard for three reasons. The bases do not match: signalling needs finite dimension and compact closure, while the communication semantics uses an infinite completion and the interventional one is measure-theoretic. The natural meeting points, parallel composition and indefinite order, are precisely where algebraic theories and fixed-order combs stop working. And the correspondence theorems, completeness and full abstraction, are each a substantial proof.

# 9 Approach

**Method.** (1) Fix a small language whose *syntax* carries the causal structure: I/O operations for communication events, labels for interventions. (2) Define an *extensional* equivalence from what an observer may do: quantify over quantum-stream environments, or over interventional contexts. (3) Build a *denotational* model in which the causal structure becomes an index, a trace or a grade, with a reindexing action. (4) Prove a *correspondence theorem*, completeness or full abstraction, between (2) and (3). The interventional route has been through all four steps; the communication route has been through (1)–(3) and has a draft of (4). Work package 1 finishes the communication route and gives it its signalling reading; work package 2 extends it to several channels; work package 3 tests the hypothesis that the two routes are one; work package 4 is the stretch beyond definite order.

**What is new.** The order of communication events is *denotational data*, an index of the model, rather than a label on an operational transition, which is what makes a signalling reading possible at all. Causal structure is *defined* by what an observer may do, a stream or a context, rather than by a property of a graph, which is what makes it compositional. And the bridge, if it holds, identifies intervening at a label with plugging a channel into a slot, giving the first semantics in which the causal-inference and quantum-foundations notions of causality are the same operation seen from two sides.

**Worked example: communication.** In $\mathsf{IO} \otimes \mathsf{Q}$, $\mathsf{in} : \langle 0 \mid 1 \rangle$ receives a qubit and binds it, $\mathsf{out} : \langle 1 \mid 0 \rangle$ sends one, and the operations of $\mathsf{Q}$ are $\mathsf{new}$, $\mathsf{measure}$ (which branches on the outcome) and $\mathsf{apply}_U$. Consider the term that receives a qubit, prepares a Bell pair locally, performs a Bell measurement on the received qubit and one half of the pair, applies the Pauli correction determined by the two outcomes to the other half, and sends it:

$$\begin{aligned} \mathit{tele} \;=\; \mathsf{in}\big(q.\ \mathsf{new}(a.\ \mathsf{new}(b.\ \mathsf{apply}_H(a,\ a.\ \mathsf{apply}_{CX}((a,b),\ (a,b).\ \\ \mathsf{apply}_{CX}((q,a),\ (q,a).\ \mathsf{apply}_H(q,\ q.\ \mathsf{measure}(q,\ c_0,\ c_1)))))))\big) \end{aligned}$$

where $c_i = \mathsf{measure}(a,\ \mathsf{apply}_{Z^i}(b,\ b.\ \mathsf{out}(b,x)),\ \mathsf{apply}_{Z^iX}(b,\ b.\ \mathsf{out}(b,x)))$. Write $T[y]$ for the local part of this term with the continuation $y(b)$ in place of each $\mathsf{out}(b,x)$; $T[y]$ is a term of $\mathsf{Q}$ alone, its denotation is the identity channel from $q$ to $b$, and Staton's theory is complete [135], so $T[y] = y(q)$ is derivable. Equational derivations are closed under substitution for continuation variables, hence $\mathit{tele} = \mathsf{in}(q.\ \mathsf{out}(q,x))$ in $\mathsf{IO}\otimes\mathsf{Q}$: the term is provably equal to a bare relay. Its denotation has a single non-zero trace component, at $\sigma = (\mathsf{in}, \mathsf{out})$, and that component is, up to the Choi correspondence, the identity on one qubit: a comb with one input slot followed by one output slot.

The derivation above uses only the local theory; the *tensor* is exhibited by the following equation from our paper, which holds in $\mathsf{IO} \otimes \mathsf{Q}$ but not in the sum of the two theories:

$$\begin{aligned} &\mathsf{new}(q.\ \mathsf{apply}_H(q,\ q.\ \mathsf{new}(p.\ \mathsf{apply}_{CX}((q,p),\ (q,p).\ \mathsf{out}(p,\ \mathsf{discard}(q,x))))))\\ &\quad = \mathsf{new}(q.\ \mathsf{apply}_H(q,\ q.\ \mathsf{measure}(q,\ \mathsf{new}(p.\ \mathsf{out}(p,x)),\ \mathsf{new}(p.\ \mathsf{apply}_X(p,\ p.\ \mathsf{out}(p,x)))))). \end{aligned}$$

Preparing a Bell pair, sending one half and discarding the other is indistinguishable, to any environment, from tossing a fair coin and sending its outcome as a classical bit. The derivation needs the commutation axiom $\mathsf{measure}(a,\ \mathsf{out}(b,x),\ \mathsf{out}(b,y)) = \mathsf{out}(b,\ \mathsf{measure}(a,x,y))$: the measurement of

the discarded half and the sending of the other half may be reordered because neither can signal to the other.

The teleportation term has one branch and one trace, which hides what makes the signalling reading non-trivial. A term that branches on a measurement, such as $\mathsf{new}(a.\ \mathsf{out}(a,\ \mathsf{in}(q.\ \mathsf{measure}(q,\ x,\ \mathsf{new}(b.\ \mathsf{out}(b$ has an $(\mathsf{out}, \mathsf{in})$-component that is post-selected on the later measurement outcome: it is sub-normalised, and on its own it is not a comb, since its trace depends on the later input. Only the family of components over all traces, with the later wires discarded, satisfies the comb conditions. The right objects are therefore testers and strategies [32, 62]:

**Conjecture 1.** *For every closed term $t$ of $\mathsf{IO} \otimes \mathsf{Q}$ and every trace $\sigma$, the component $[\![t]\!]_\sigma$ is a probabilistic comb with slot order $\sigma$, and the family $([\![t]\!]_\sigma)_\sigma$ is a variable-length tester: for each $\sigma$, the sum of the components over all extensions of $\sigma$, with the later wires discarded, is a deterministic comb with slot order $\sigma$.*

**Worked example: the comparison target.** In $L_2$, interventions are exposed as typed free variables $\ell$ read through applyIntv, which returns the label's value when it is set and the program's own value otherwise:

```
let t  = bernoulli 0.5 in              -- treatment
let t' = applyIntv ℓ t in             -- intervenable copy of t
let m  = bernoulli (f t') in           -- mediator
let y  = bernoulli (g t' m) in         -- outcome
let z  = bernoulli 0.3 in              -- unrelated variable
return (y, z)
```

Under $\mathsf{do}(\ell \mapsto \mathsf{Just\ true})\{-\}$ the variable $t'$ denotes the constant true, and the total effect of $t$ on $y$ is the difference between the distributions of $y$ under the two perfect interventions. The variable $z$ has the same distribution in every interventional context, and $\equiv_2$ makes this an equation between programs of the same type: the program with `return` $z$ in place of `return` $(y, z)$ is $\equiv_2$ to `let` $t' =$ `applyIntv` $\ell$ `true in let` $z =$ `bernoulli` $0.3$ `in return` $z$, in which the label is consumed but ignored. Full abstraction says that the graded denotation detects exactly these equations. Read side by side with the Bell-pair equation, this is the same statement: no information flows from one place to another, so intervening or reordering is unobservable. The hypothesis of WP 3 says why.

**Work package 1: the signalling reading of single-channel communication.**

1a *(low risk)* Close and submit the completeness proof: the model of $\mathsf{IO} \otimes \mathsf{Q}$ in $\mathrm{Kl}(T)^{\mathrm{op}}$ is complete, by induction on I/O depth with a reorganisation step and a factorisation lemma on CP maps. *Done:* the two open steps of the draft discharged; manuscript submitted. With it, the algebraic, operational and denotational semantics coincide.

1b *(low risk)* Conjecture 1. *Done:* proved by structural induction, with the normalisation bookkeeping across traces of different lengths made explicit. *Why it may succeed:* sequencing is comb composition, and the in/out directions are exactly the comb's input and output wires.

1c *(medium risk)* The typing functor and relative completeness.

**Conjecture 2.** *For each trace $\sigma$, the component $[\![t]\!]_\sigma$ lands in the comb type for $\sigma$ in* **Caus**(**CP**)*, the branch family is typed by the coproduct and probabilistic-choice operators of the additive construction [129], and the assignment is functorial in sequencing. Every deterministic comb of finite depth with qubit slots is the component of some term, and the equations of $\mathsf{IO} \otimes \mathsf{Q}$ are exactly those validated by the comb types.*

The first sentence is the typing statement; the second is expected to hold by the comb realisation theorem [32] together with the universality of $\mathsf{Q}$; the third, completeness of $\mathsf{IO} \otimes \mathsf{Q}$ relative to **Caus**, is the open question, and causal logic [129, 131] gives it its sharp form: which fragment of causal logic do the I/O orders of $\mathsf{IO} \otimes \mathsf{Q}$ terms realise, and is every causally consistent wiring in that fragment denotable? The trace index is kept external, so the construction is over finite-dimensional **CP** and not over $\mathbf{CP}^\infty$. *Done:* the functor constructed;

realisation proved or refuted; relative completeness answered or reduced to a precisely stated question. *Why it may succeed:* the objects on both sides are finite-dimensional CP maps with the same wire structure, and the completeness proof of 1a already produces the normal forms that a comparison with comb types needs.

1d *(medium risk)* A contextual equivalence for $\mathsf{IO} \otimes \mathsf{Q}$. The operational equivalence $\approx_q$ inspects final quantum states and is not contextual; the question is whether the equivalence under all contexts that observe only traces, continuations and a final measurement coincides with it. *Done:* a coincidence theorem or a separating pair of terms. *Why it matters:* it is what makes step (2) of the method literally the same on both routes, and it is the form in which WP 3 is stated.

**Work package 2: several channels.**

2a *(medium risk; in the minimum viable thesis)* Independent channels in a sequential program: $\mathsf{IO}\otimes\mathsf{Q}$ with operations $\mathsf{in}_c$, $\mathsf{out}_c$ for each channel $c$, where events on different channels commute, so that the monad is indexed by Mazurkiewicz traces [45] over a partially commutative monoid. *Done:* the theory defined, non-degeneracy by a multi-stream operational model, soundness and adequacy of the trace-indexed model; completeness by re-running the single-channel proof, whose reorganisation step and factorisation lemma are channel-agnostic. *Why it may succeed:* the proof architecture is unchanged; what is new is the commutation of independent I/O events, which the quantum-stream model must validate. *Test:* teleportation as a two-party program; entanglement swapping. The signalling reading of 1c extends to graph types [129], since a Mazurkiewicz trace is a DAG of events.

2b *(high risk; stretch)* Threads: tensor $\mathsf{IO} \otimes \mathsf{Q}$ with the dynamic-threads theory of Kammar et al. [76] and ask which of its equations survive the quantum setting, in the manner of the Eckmann–Hilton analysis in my dissertation; traces become pomsets. *Done:* the theory defined and shown non-degenerate; a soundness theorem. Simmons's conjectured correspondence between causal consistency and deadlock freedom [129] is the natural statement to aim at once threads are present.

**Work package 3: the bridge—interventions as communication.**

**Hypothesis 1.** *On the finite classical fragment, an intervention label is an* $\mathsf{out}/\mathsf{in}$ *pair with the intervener as the environment: the program sends the natural value and receives the value it will use. The interventional context* $\mathsf{do}(\ell \mapsto v)$ *is the stream that ignores the output and returns* $v$*; "no intervention" is the echo stream; and* $\equiv_2$ *is operational equivalence under the class of memoryless echo-or-replace streams.*

If Hypothesis 1 holds, the interventional route is literally a fragment of the communication route: $L_2$ programs are $\mathsf{IO} \otimes \mathsf{Q}$-style programs over **FinStoch** instead of **CP**, and the two existing full-abstraction theorems are two instances of one statement. Why it should hold: $\mathsf{applyInt}\,\ell\,t$ outputs $t$ and inputs the label's value or $t$, which is exactly what an echo-or-replace stream does to an $\mathsf{out}$ followed by an $\mathsf{in}$; the content is in showing that the contextual quantification over labels matches the quantification over streams, and that nothing is lost by restricting to memoryless streams.

3a *(medium risk; in the minimum viable thesis)* Prove or refute Hypothesis 1; then construct the typing functor from the graded interventional category over **FinStoch** into $\mathbf{Caus}(\mathbf{Mat}(\mathbb{R}_{\geq 0}))$, sending intervention labels to slots and grades to signalling types, compatible with composition. Two refinements are built in. Perfect and soft interventions differ: invariance of $Y$ under all perfect interventions at a slot does not imply invariance under all soft ones, whereas invariance under all soft interventions is exactly no-signalling from the slot; the theorem is stated for soft interventions. And a *factorisation* reading is included: the conditional independences of the Markov-category image, via Fritz and Klingler's synthetic d-separation [53], as the third face of the same functor. The higher-order fragment is obstructed by sub-normalisation from conditioning, so the functor is first defined on the normalised fragment and the extension to

testers is a stated sub-question. *Done:* the hypothesis settled; the functor and its properties, or a counterexample program; the soft/perfect separation; the factorisation reading. Either outcome is a result. Comb disintegration [74] is the prior work this item must be positioned against; it uses combs to compute with a fixed classical model, whereas here the comb type is assigned to a program and compared with its contextual equivalence.

3b *(risk unknown until milestone M2)* Quantum causal models [12] in the same setting over finite-dimensional **CP**, with nodes as labels and graph types [129] for DAG-shaped signalling. *Done:* the quantum Markov condition recovered as a property of the trace-indexed model, or a precise statement of what is missing.

**Work package 4 (stretch): beyond definite order.**

4a *(high risk; conditional on 2a and 1c)* Networks of communicating programs as process matrices: local programs are combs, the network that wires them is a process matrix, and the classical-control and quantum-control classes of Wechs et al. [143] are the candidate denotational universe for the network component. *Done:* a translation from parallel compositions under a classical scheduler into process matrices with its image shown to lie in the classical-control class, hence causally separable; the converse, that every classical-control process arises this way, proved or refuted; a catalogue of protocols (teleportation, entanglement swapping, a BB84 fragment, a two-party comb) with denotations matching the known process operators. Realisability cannot be a **Caus** type, since every process matrix is an affine combination of definite-order ones [129]; it is a predicate on denotations. [FILL: the Kim reference for this item]

4b *(high risk; exploratory)* Generalised switches as quantum schedulers over the linear extensions of a pomset of events; realisability is expected as a corollary of the $N$-party switch, and the questions with content are which quantum-control processes arise, what a quantum-scheduler primitive is in a language whose continuations are classical, in relation to the PBS-calculus [38], and the condition that the composed programs share a target.

**Not pursued.** The interventional route's own next steps, its equational theory against normalisation boxes and cyclic models as fixpoints, are natural but are not part of this thesis; the equational-theory item may fall out of WP 3a as a by-product.

## 10 Evaluation

The thesis is theoretical, and its results are evaluated as theorems with the "done" criteria above: completeness, the tester and typing statements, the bridge hypothesis, and the coincidence or separation results of WP 3a. Case studies have pass/fail criteria: for the protocol catalogue, each protocol either has a term whose denotation is the known process operator or a proof that none does; for the interventional comparison, the mediation example and its relatives are reproduced through the bridge and give the same answers as in the interventional semantics. The executable semantics [139] will be extended with the communication constructs so that every definition remains executable.

## 11 Impact and risks

If successful, the thesis gives the first denotational semantics of communicating quantum programs in which signalling structure is typed and checked, a semantic model of distributed quantum programming in which realisability as a controlled-order circuit is a predicate on denotations, and, through the bridge, a precise sense in which the causal-inference and quantum-foundations notions of causality are one.

The generic risk is reaching into higher-order quantum theory from a programming-language background. The mitigation is that the baseline, the *minimum viable thesis*, needs only the Caus

| Item | Risk | Fallback |
|---|---|---|
| 1a completeness | low | the adequacy and full-abstraction results stand |
| 1b tester reading (B1) | low | — |
| 1c typing and relative completeness (B2) | medium | the typing functor and realisation alone characterise the fragment |
| 1d contextual equivalence | medium | a separating pair is a result |
| 2a independent channels | medium | soundness and adequacy without completeness |
| 2b threads | high, stretch | dropped without loss |
| 3a bridge | medium | a refutation of the hypothesis is a publishable separation |
| 3b quantum causal models | unknown until M2 | a precise statement of what is missing |
| 4a networks, 4b switches | high, stretch | dropped without loss |

Table 2: Risks and fallbacks.

construction, combs and Simmons's chapter 2: it consists of WP 1, WP 2a and WP 3a, every item rated low or medium, with the interventional chapter as finished work. The process-matrix literature is needed only by WP 4.

## 12 Plan of work, milestones, and training

I am in my fourth term. The baseline schedule is the minimum viable thesis; conditional items are marked.

| Term | Milestones |
|---|---|
| MT26 (4) | Transfer of status. POPL revision. Reading milestone M1: Simmons's thesis [129], chapters 2–3; Kammar et al. [76]; the papers of §4.4–4.5 at the level of their abstracts. |
| HT27 (5) | WP 1a closed and submitted. WP 1b proved. Hypothesis 1 checked on paper (WP 3a, first half). |
| TT27 (6) | WP 1c: typing functor and realisation. WP 3a: the functor. *Deliverable:* paper on completeness, the tester reading and the bridge. |
| MT27–HT28 (7–8) | WP 2a. WP 1d. Reading milestone M2: quantum causal models [12]. *Deliverable:* paper on independent channels. **Confirmation of Status in HT28**, with the minimum viable thesis in hand. |
| TT28 (9) | Buffer for Confirmation of Status. WP 3b. Relative completeness of WP 1c if not yet settled. |
| MT28 (10) | WP 4a if 2a and 1c are complete; otherwise WP 2b. |
| HT29–TT29 (11–12) | WP 4b only if 4a is done; otherwise consolidation. Thesis writing; submission by the end of TT29. |

Dissemination targets are POPL, LICS, FoSSaCS, MFPS and QPL. The thesis will be organised as: 1 Introduction; 2 Background (Part I of this document); 3 The interventional template: causal probabilistic programs [FILL: delineate your contribution from coauthors']; 4 Order semantics of quantum communication: completeness and the tester reading, with the MSc dissertation as background and not as DPhil material [FILL: confirm reuse rules with supervisor]; 5 Several channels; 6 The bridge; 7 Beyond definite order (if WP 4 lands); 8 Conclusion.

**Training and engagement.** [FILL: required by the assessors' criteria: seminars attended (PL group, quantum group); courses audited; the PLanQC 2025 talk, with a paragraph on it; teaching; planned activities for the coming year].

## 13 Statement of the subject of the thesis

A semantics of order and signalling for communicating quantum programs, with interventional causality as its classical fragment.

## References


[1] M. Abadi and G. D. Plotkin. "A Model of Cooperative Threads". In: *Log. Methods Comput. Sci.* 6.4 (2010). DOI: 10.2168/LMCS-6(4:2)2010. URL: https://doi.org/10.2168/LMCS-6(4:2)2010.

[2] A. A. Abbott, J. Wechs, D. Horsman, M. Mhalla, and C. Branciard. "Communication through coherent control of quantum channels". In: *Quantum* 4 (Sept. 2020), p. 333. ISSN: 2521-327X. DOI: 10.22331/q-2020-09-24-333. URL: http://dx.doi.org/10.22331/q-2020-09-24-333.

[3] S. Abramsky and B. Coecke. "A categorical semantics of quantum protocols". In: *Proceedings of the 19th Annual IEEE Symposium on Logic in Computer Science, LICS '04*. 2004, pp. 415–425. DOI: 10.1109/LICS.2004.1319636.

[4] O. Abril-Pla et al. "PyMC: a modern, and comprehensive probabilistic programming framework in Python". In: *PeerJ Computer Science* 9 (2023), e1516. DOI: 10.7717/peerj-cs.1516.

[5] S. Agrawal, J. Bao, and S. Holtzen. "Towards an Equational Calculus of Interventions". In: *Languages for Inference (LAFI)*. 2026.

[6] D. Ahman and A. Bauer. "Runners in Action". In: *Programming Languages and Systems*. Ed. by P. Müller. Cham: Springer International Publishing, 2020, pp. 29–55. ISBN: 978-3-030-44914-8.

[7] J.-M. A. Allen, J. Barrett, D. C. Horsman, C. M. Lee, and R. W. Spekkens. "Quantum common causes and quantum causal models". In: *Physical Review X* 7 (2017), p. 031021.

[8] T. Altenkirch and A. S. Green. "The Quantum IO Monad". In: *Semantic Techniques in Quantum Computation*. Ed. by S. Gay and I. Mackie. Cambridge University Press, 2009, pp. 173–205.

[9] M. Araújo, C. Branciard, F. Costa, A. Feix, C. Giarmatzi, and Č. Brukner. "Witnessing causal nonseparability". In: *New Journal of Physics* 17 (2015), p. 102001.

[10] A. Balke and J. Pearl. "Probabilistic evaluation of counterfactual queries". In: *Proceedings of the Twelfth AAAI National Conference on Artificial Intelligence*. AAAI'94. Seattle, Washington: AAAI Press, 1994, pp. 230–237.

[11] E. Bareinboim, J. D. Correa, D. Ibeling, and T. Icard. "On Pearl's Hierarchy and the Foundations of Causal Inference". In: *Probabilistic and Causal Inference: The Works of Judea Pearl*. 1st. New York, NY, USA: Association for Computing Machinery, 2022, pp. 507–556.

[12] J. Barrett, R. Lorenz, and O. Oreshkov. "Quantum causal models". In: *arXiv preprint arXiv:1906.10726* (2019).

[13] J. Barrett, R. Lorenz, and O. Oreshkov. "Cyclic quantum causal models". In: *Nature Communications* 12.885 (2021). DOI: 10.1038/s41467-020-20456-x.

[14] A. Bauer. *What is algebraic about algebraic effects and handlers?* 2019. arXiv: 1807.05923 [cs.LO]. URL: https://arxiv.org/abs/1807.05923.

[15] M. R. Becker et al. "Probabilistic Programming with Vectorized Programmable Inference". In: *Proc. ACM Program. Lang.* 10.POPL (Jan. 2026). DOI: 10.1145/3776729. URL: https://doi.org/10.1145/3776729.

[16] S. Beckers and J. Y. Halpern. "Abstracting causal models". In: *Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence and Thirty-First Innovative Applications of Artificial Intelligence Conference and Ninth AAAI Symposium on Educational Advances in Artificial Intelligence*. AAAI'19/IAAI'19/EAAI'19. Honolulu, Hawaii, USA: AAAI Press, 2019. ISBN: 978-1-57735-809-1. DOI: 10.1609/aaai.v33i01.33012678. URL: https://doi.org/10.1609/aaai.v33i01.33012678.

[17] C. H. Bennett and G. Brassard. "Quantum cryptography: Public key distribution and coin tossing". In: *Theoretical Computer Science* 560 (2014). Theoretical Aspects of Quantum Cryptography – celebrating 30 years of BB84, pp. 7–11. ISSN: 0304-3975. DOI: 10.1016/j.tcs.2014.05.025. URL: https://www.sciencedirect.com/science/article/pii/S0304397514004241.

[18] C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres, and W. K. Wootters. "Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels". In: *Phys. Rev. Lett.* 70 (13 Mar. 1993), pp. 1895–1899. DOI: 10.1103/PhysRevLett.70.1895. URL: https://link.aps.org/doi/10.1103/PhysRevLett.70.1895.

[19] B. Bichsel, M. Baader, T. Gehr, and M. Vechev. "Silq: a high-level quantum language with safe uncomputation and intuitive semantics". In: *Proceedings of the 41st ACM SIGPLAN Conference on Programming Language Design and Implementation*. PLDI 2020. London, UK: Association for Computing Machinery, 2020, pp. 286–300. ISBN: 9781450376136. DOI: 10.1145/3385412.3386007. URL: https://doi.org/10.1145/3385412.3386007.

[20] E. Bingham et al. "Pyro: Deep Universal Probabilistic Programming". In: *Journal of Machine Learning Research* (2018).

[21] A. Bisio and P. Perinotti. "Theoretical framework for higher-order quantum theory". In: *Proceedings of the Royal Society A* 475 (2019), p. 20180706.

[22] P. Blöbaum, P. Götz, K. Budhathoki, A. A. Mastakouri, and D. Janzing. "DoWhy-GCM: An extension of DoWhy for causal inference in graphical causal models". In: *Journal of Machine Learning Research* 25 (2024). URL: https://jmlr.org/papers/v25/22-1258.html.

[23] S. Bongers, P. Forré, J. Peters, and J. M. Mooij. "Foundations of structural causal models with cycles and latent variables". In: *The Annals of Statistics* 49.5 (Oct. 2021). ISSN: 0090-5364. DOI: 10.1214/21-aos2064. URL: http://dx.doi.org/10.1214/21-AOS2064.

[24] J. Brulé. *Whittemore: An embedded domain specific language for causal programming*. 2018. arXiv: 1812.11918 [cs.AI]. URL: https://arxiv.org/abs/1812.11918.

[25] S. Buchholz, J. Park, and B. Schölkopf. "Products, abstractions and inclusions of causal spaces". In: *arXiv preprint arXiv:2406.00388* (2024).

[26] M. Caleffi, M. Amoretti, D. Ferrari, J. Illiano, A. Manzalini, and A. S. Cacciapuoti. "Distributed quantum computing: A survey". In: *Computer Networks* 254 (Dec. 2024), p. 110672. ISSN: 1389-1286. DOI: 10.1016/j.comnet.2024.110672. URL: http://dx.doi.org/10.1016/j.comnet.2024.110672.

[27] J. Carette, C. Heunen, R. Kaarsgaard, and A. Sabry. "With a Few Square Roots, Quantum Computing Is as Easy as Pi". In: *Proceedings of the ACM on Programming Languages* 8.POPL (Jan. 2024), pp. 546–574. ISSN: 2475-1421. DOI: 10.1145/3632861. URL: http://dx.doi.org/10.1145/3632861.

[28] L. Ceragioli, E. Di Lavore, G. Lomurno, and G. Tedeschi. "A Coalgebraic Model of Quantum Bisimulation". In: *Proceedings of the Seventh International Conference on Applied Category Theory (ACT 2024)*. Vol. 429. EPTCS. 2025, pp. 249–269. DOI: 10.4204/EPTCS.429.14.

[29] L. Ceragioli, F. Gadducci, G. Lomurno, and G. Tedeschi. "Quantum Bisimilarity via Barbs and Contexts: Curbing the Power of Non-deterministic Observers". In: *Proc. ACM Program. Lang.* 8.POPL (Jan. 2024). DOI: 10.1145/3632885. URL: https://doi.org/10.1145/3632885.

[30] *ChiRho*. https://basisresearch.github.io/chirho. Accessed: 2025-10-30.

[31] G. Chiribella, G. M. D'Ariano, and P. Perinotti. "Quantum Circuit Architecture". In: *Physical Review Letters* 101.6 (Aug. 2008). ISSN: 1079-7114. DOI: 10.1103/physrevlett.101.060401. URL: http://dx.doi.org/10.1103/PhysRevLett.101.060401.

[32] G. Chiribella, G. M. D'Ariano, and P. Perinotti. "Theoretical framework for quantum networks". In: *Physical Review A* 80 (2009), p. 022339.

[33] G. Chiribella, G. M. D'Ariano, and P. Perinotti. "Probabilistic theories with purification". In: *Physical Review A* 81 (2010), p. 062348.

[34] G. Chiribella, G. M. D'Ariano, P. Perinotti, and B. Valiron. "Quantum computations without definite causal structure". In: *Physical Review A* 88 (2013), p. 022318. DOI: 10.1103/PhysRevA.88.022318.

[35] K. Cho. "Semantics for a Quantum Programming Language by Operator Algebras". In: *New Generation Computing* 34.1 (2016), pp. 25–68. DOI: 10.1007/s00354-016-0204-3. URL: https://doi.org/10.1007/s00354-016-0204-3.

[36] K. Cho and B. Jacobs. "Disintegration and Bayesian inversion both abstractly and concretely". In: *Mathematical Structures in Computer Science* 29.7 (2019), pp. 938–971.

[37] P. Clairambault, M. de Visme, and G. Winskel. "Game semantics for quantum programming". In: *Proceedings of the ACM on Programming Languages* 3.POPL (2019), 32:1–32:29.

[38] A. Clément and S. Perdrix. "PBS-calculus: a graphical language for coherent control of quantum computations". In: *MFCS 2020*. Vol. 170. LIPIcs. 2020, 24:1–24:14.

[39] B. Coecke. "Terminality implies no-signalling". In: *Proceedings of QPL 2014*. Vol. 172. EPTCS. 2014, pp. 27–35.

[40] B. Coecke and A. Kissinger. *Picturing Quantum Processes: A First Course in Quantum Theory and Diagrammatic Reasoning*. Cambridge University Press, 2017.

[41] F. Costa and S. Shrapnel. "Quantum causal modelling". In: *New Journal of Physics* 18 (2016), p. 063032.

[42] S. Dash, Y. Kaddar, H. Paquet, and S. Staton. "Affine Monads and Lazy Structures for Bayesian Programming". In: *Proc. ACM Program. Lang.* 7.POPL (Jan. 2023). DOI: 10.1145/3571239. URL: https://doi.org/10.1145/3571239.

[43] A. P. Dawid. "Influence Diagrams for Causal Modelling and Inference". In: *International Statistical Review* 70.2 (2002), pp. 161–189. DOI: https://doi.org/10.1111/j.1751-5823.2002.tb00354.x.

[44] V. Didelez, A. P. Dawid, and S. Geneletti. "Direct and indirect effects of sequential treatments". In: *Proceedings of the Twenty-Second Conference on Uncertainty in Artificial Intelligence*. UAI'06. Cambridge, MA, USA: AUAI Press, 2006, pp. 138–146. ISBN: 0974903922.

[45] V. Diekert and G. Rozenberg, eds. *The Book of Traces*. World Scientific, 1995.

[46] U. Dorsch, S. Milius, and L. Schröder. "Graded Monads and Graded Logics for the Linear Time–Branching Time Spectrum". In: *CONCUR 2019* (2019).

[47] Y. Dvir, O. Kammar, and O. Lahav. "An algebraic theory for shared-state concurrency". In: *Programming Languages and Systems (APLAS 2022)*. Vol. 13658. LNCS. 2022, pp. 3–24. DOI: 10.1007/978-3-031-21037-2_1.

[48] F. Eberhardt and R. Scheines. "Interventions and Causal Inference". In: *Philosophy of Science* 74.5 (2007), pp. 981–995. DOI: 10.1086/525638.

[49] T. Ehrhard, M. Pagani, and C. Tasson. "Probabilistic coherence spaces are fully abstract for probabilistic PCF". In: *POPL 2014*. 2014, pp. 309–320.

[50] M. Englberger and D. S. Dhami. *Causal Abstractions, Categorically Unified*. 2025. arXiv: 2510.05033 [stat.ML]. URL: https://arxiv.org/abs/2510.05033.

[51] B. Fong. "Causal theories: a categorical perspective on Bayesian networks". arXiv:1301.6201. MA thesis. University of Oxford, 2012.

[52] T. Fritz. "A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics". In: *Advances in Mathematics* 370 (2020), p. 107239.

[53] T. Fritz and A. Klingler. "The d-separation criterion in categorical probability". In: *Journal of Machine Learning Research* 24.46 (2023), pp. 1–49.

[54] T. Fritz and P. Perrone. "A probability monad as the colimit of spaces of finite samples". In: *Theory and Applications of Categories* (2019).

[55] C. Führmann. "Direct models of the computational lambda-calculus". In: *Electronic Notes in Theoretical Computer Science* 20 (1999), pp. 245–292.

[56] S. Fujii, S.-y. Katsumata, and P.-A. Melliès. "Towards a formal theory of graded monads". In: *International Conference on Foundations of Software Science and Computation Structures*. Springer. 2016, pp. 513–530.

[57] S. Gay and R. Nagarajan. "Communicating Quantum Processes". In: *Proceedings of the 32nd ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages, POPL '05*. 2005, pp. 145–157. DOI: 10.1145/1040305.1040318.

[58] J. L. Gischer. "The equational theory of pomsets". In: *Theoretical Computer Science* 61.2–3 (1988), pp. 199–224.

[59] R. van Glabbeek and G. Plotkin. "On CSP and the Algebraic Theory of Effects". In: *Reflections on the Work of C.A.R. Hoare*. Springer London, 2010, pp. 333–369. ISBN: 9781848829121. DOI: 10.1007/978-1-84882-912-1_15. URL: http://dx.doi.org/10.1007/978-1-84882-912-1_15.

[60] J. A. Goguen and J. Meseguer. "Security policies and security models". In: *IEEE Symposium on Security and Privacy*. 1982, pp. 11–20.

[61] A. S. Green, P. L. Lumsdaine, N. J. Ross, P. Selinger, and B. Valiron. "Quipper: A Scalable Quantum Programming Language". In: *CoRR* abs/1304.3390 (2013). arXiv: 1304.3390. URL: http://arxiv.org/abs/1304.3390.

[62] G. Gutoski and J. Watrous. "Toward a general theory of quantum games". In: *STOC 2007*. 2007, pp. 565–574.

[63] J. Y. Halpern. *Actual Causality*. MIT Press, 2016.

[64] C. Heunen, O. Kammar, S. Staton, and H. Yang. "A convenient category for higher-order probability theory". In: *2017 32nd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*. IEEE, June 2017, pp. 1–12. DOI: 10.1109/lics.2017.8005137. URL: http://dx.doi.org/10.1109/LICS.2017.8005137.

[65] C. Heunen and J. Vicary. *Categories for Quantum Theory: An Introduction*. Oxford University Press, Nov. 2019. ISBN: 9780198739623. DOI: 10.1093/oso/9780198739623.001.0001. eprint: https://academic.oup.com/book/43710/book-pdf/50991591/9780191060069_web.pdf. URL: https://doi.org/10.1093/oso/9780198739623.001.0001.

[66] C. A. R. Hoare. "Communicating sequential processes". In: *Commun. ACM* 21.8 (Aug. 1978), pp. 666–677. ISSN: 0001-0782. DOI: 10.1145/359576.359585. URL: https://doi.org/10.1145/359576.359585.

[67] T. Hoare, B. Möller, G. Struth, and I. Wehrman. "Concurrent Kleene algebra and its foundations". In: *Journal of Logic and Algebraic Programming* 80.6 (2011), pp. 266–296.

[68] M. Hyland, G. Plotkin, and J. Power. "Combining effects: Sum and tensor". In: *Theoretical Computer Science* 357.1 (2006). Clifford Lectures and the Mathematical Foundations of Programming Semantics, pp. 70–99. ISSN: 0304-3975. DOI: 10.1016/j.tcs.2006.03.013. URL: https://www.sciencedirect.com/science/article/pii/S0304397506002659.

[69] M. Hyland and J. Power. "The Category Theoretic Understanding of Universal Algebra: Lawvere Theories and Monads". In: *Electronic Notes in Theoretical Computer Science* 172 (2007). Computation, Meaning, and Logic: Articles dedicated to Gordon Plotkin, pp. 437–458. ISSN: 1571-0661. DOI: 10.1016/j.entcs.2007.02.019. URL: https://www.sciencedirect.com/science/article/pii/S1571066107000874.

[70] D. Ibeling and T. Icard. "On Open-Universe Causal Reasoning". In: *Proceedings of The 35th Uncertainty in Artificial Intelligence Conference*. Ed. by R. P. Adams and V. Gogate. Vol. 115. Proceedings of Machine Learning Research. PMLR, 22–25 Jul 2020, pp. 1233–1243. URL: https://proceedings.mlr.press/v115/ibeling20a.html.

[71] D. Ibeling and T. Icard. "Probabilistic Reasoning across the Causal Hierarchy". In: *CoRR* abs/2001.02889 (2020). arXiv: 2001.02889. URL: http://arxiv.org/abs/2001.02889.

[72] D. Ibeling and T. Icard. "Comparing causal frameworks: potential outcomes, structural models, graphs, and abstractions". In: *Proceedings of the 37th International Conference on Neural Information Processing Systems*. NIPS '23. New Orleans, LA, USA: Curran Associates Inc., 2023.

[73] T. F. Icard. "From Programs to Causal Models". In: *Proc. 21st Amsterdam Colloquium*. 2017.

[74] B. Jacobs, A. Kissinger, and F. Zanasi. "Causal inference by string diagram surgery". In: *Foundations of Software Science and Computation Structures: 22nd International Conference, FOSSACS 2019, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2019, Prague, Czech Republic, April 6–11, 2019, Proceedings 22*. Springer. 2019, pp. 313–329.

[75] B. Jacobs, M. Széles, and D. Stein. "Compositional Inference for Bayesian Networks and Causality". In: *Electronic Notes in Theoretical Informatics and Computer Science* Volume 5 - Proceedings of MFPS XLI, 11 (Dec. 2025). ISSN: 2969-2431. DOI: 10.46298/entics.17029. URL: https://entics.episciences.org/17029.

[76] O. Kammar, J. Liell-Cock, S. Lindley, C. Matache, and S. Staton. "An equational axiomatization of dynamic threads via algebraic effects". In: *Proceedings of the ACM on Programming Languages* 10.POPL (2026). arXiv:2602.05850, pp. 1847–1875.

[77] T. Kappé, P. Brunet, A. Silva, and F. Zanasi. "Concurrent Kleene algebra: free model and completeness". In: *Programming Languages and Systems (ESOP 2018)*. Vol. 10801. LNCS. 2018, pp. 856–882. DOI: 10.1007/978-3-319-89884-1_30.

[78] S.-y. Katsumata. "Parametric effect monads and semantics of effect systems". In: *SIGPLAN Not.* 49.1 (Jan. 2014), pp. 633–645. ISSN: 0362-1340. DOI: 10.1145/2578855.2535846. URL: https://doi.org/10.1145/2578855.2535846.

[79] R. Kiesel, K. Rückschloß, and F. Weitkämper. ""What if?" in probabilistic logic programming". In: *Theory and Practice of Logic Programming* 23.4 (2023).

[80] A. Kissinger and S. Uijlen. "A categorical semantics for causal structure". In: *Logical Methods in Computer Science* 15.3 (2019). DOI: 10.23638/LMCS-15(3:5)2019.

[81] A. Kock. "Strong functors and monoidal monads". In: *Archiv der Mathematik* 23.1 (1972), pp. 113–120. DOI: 10.1007/BF01304852. URL: https://doi.org/10.1007/BF01304852.

[82] H. Kristjánsson, G. Chiribella, S. Salek, D. Ebler, and M. Wilson. "Resource theories of communication". In: *New Journal of Physics* 22.7 (July 2020), p. 073014. DOI: 10.1088/1367-2630/ab8ef7. URL: https://dx.doi.org/10.1088/1367-2630/ab8ef7.

[83] M. Lalire and P. Jorrand. *A Process Algebraic Approach to Concurrent and Distributed Quantum Computation: Operational Semantics*. 2004. arXiv: quant-ph/0407005 [quant-ph]. URL: https://arxiv.org/abs/quant-ph/0407005.

[84] L. Lamport. "Time, clocks, and the ordering of events in a distributed system". In: *Communications of the ACM* 21.7 (1978), pp. 558–565.

[85] M. R. Laurence and G. Struth. "Completeness theorems for bi-Kleene algebras and series-parallel rational pomset languages". In: *Relational and Algebraic Methods in Computer Science (RAMiCS 2014)*. Vol. 8428. LNCS. 2014, pp. 65–82.

[86] A. W. Levis, E. H. Kennedy, A. McClean, S. Balakrishnan, and L. Wasserman. *Stochastic interventions, sensitivity analysis, and optimal transport*. 2024. arXiv: 2411.14285 [stat.ME]. URL: https://arxiv.org/abs/2411.14285.

[87] P. B. Levy. *Locally graded categories*. Slides available from the author's webpage. 2019.

[88] J. Liell-Cock and S. Staton. "Compositional imprecise probability: A solution from graded monads and markov categories". In: *Proceedings of the ACM on Programming Languages* 9.POPL (2025), pp. 1596–1626.

[89] R. Lorenz and S. Tull. *Causal models in string diagrams*. 2023. arXiv: 2304.07638 [cs.LO]. URL: https://arxiv.org/abs/2304.07638.

[90] R. Lorenz and S. Tull. *Causal and Compositional Abstraction*. 2026. arXiv: 2602.16612 [cs.LO]. URL: https://arxiv.org/abs/2602.16612.

[91] J.-W. van de Meent, B. Paige, H. Yang, and F. Wood. *An Introduction to Probabilistic Programming*. 2021. arXiv: 1809.10756 [stat.ML]. URL: https://arxiv.org/abs/1809.10756.

[92] S. Milius, D. Pattinson, and L. Schröder. "Generic trace semantics and graded monads". In: *CALCO 2015*. Vol. 35. LIPIcs. 2015, pp. 253–269.

[93] R. Milner. *A Calculus of Communicating Systems*. Berlin, Heidelberg: Springer-Verlag, 1982. ISBN: 0387102353.

[94] R. Milner, J. Parrow, and D. Walker. "A calculus of mobile processes, I". In: *Information and Computation* 100.1 (1992), pp. 1–40. ISSN: 0890-5401. DOI: 10.1016/0890-5401(92)90008-4. URL: https://www.sciencedirect.com/science/article/pii/0890540192900084.

[95] E. Moggi. "Notions of computation and monads". In: *Information and Computation* 93.1 (1991). Selections from 1989 IEEE Symposium on Logic in Computer Science, pp. 55–92. ISSN: 0890-5401. DOI: 10.1016/0890-5401(91)90052-4. URL: https://www.sciencedirect.com/science/article/pii/0890540191900524.

[96] E. Moggi. "Notions of computation and monads". In: *Inf. Comput.* 93.1 (July 1991), pp. 55–92. ISSN: 0890-5401. DOI: 10.1016/0890-5401(91)90052-4. URL: https://doi.org/10.1016/0890-5401(91)90052-4.

[97] M. Nielsen, G. Plotkin, and G. Winskel. "Petri nets, event structures and domains, part I". In: *Theoretical Computer Science* 13.1 (1981), pp. 85–108.

[98] D. Orchard, P. Wadler, and H. Eades III. "Unifying graded and parameterised monads". In: *Proc. MSFP 2020* (2020).

[99] O. Oreshkov. "Time-delocalized quantum subsystems and operations: on the existence of processes with indefinite causal structure in quantum mechanics". In: *Quantum* 3 (2019), p. 206.

[100] O. Oreshkov, F. Costa, and Č. Brukner. "Quantum correlations with no causal order". In: *Nature Communications* 3.1092 (2012). DOI: 10.1038/ncomms2076.

[101] O. Oreshkov and C. Giarmatzi. "Causal and causally separable processes". In: *New Journal of Physics* 18 (2016), p. 093020.

[102] M. Pagani, P. Selinger, and B. Valiron. "Applying quantitative semantics to higher-order quantum computing". In: *Proceedings of the 41st ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages, POPL '14*. 2014, pp. 647–658. DOI: 10.1145/2535838.2535879.

[103] J. Park, S. Buchholz, B. Schölkopf, and K. Muandet. "A measure-theoretic axiomatisation of causality". In: *Proceedings of the 37th International Conference on Neural Information Processing Systems*. NIPS '23. New Orleans, LA, USA: Curran Associates Inc., 2023.

[104] J. Paykin, R. Rand, and S. Zdancewic. "QWIRE: a core language for quantum circuits". In: *Proceedings of the 44th ACM SIGPLAN Symposium on Principles of Programming Languages*. POPL '17. Paris, France: Association for Computing Machinery, 2017, pp. 846–858. ISBN: 9781450346603. DOI: 10.1145/3009837.3009894. URL: https://doi.org/10.1145/3009837.3009894.

[105] J. Pearl. *Causality*. Cambridge University Press, 2009.

[106] J. Pearl and D. Mackenzie. *The Book of Why: The New Science of Cause and Effect*. 1st. USA: Basic Books, Inc., 2018. ISBN: 046509760X.

[107] Y. Perov et al. "Multiverse: causal reasoning using importance sampling in probabilistic programming". In: *Symposium on advances in approximate bayesian inference*. PMLR. 2020, pp. 1–36.

[108] J. Peters, D. Janzing, and B. Schölkopf. *Elements of Causal Inference: Foundations and Learning Algorithms*. The MIT Press, 2017. ISBN: 0262037319.

[109] G. Plotkin and J. Power. "Notions of Computation Determine Monads". In: *Foundations of Software Science and Computation Structures*. Ed. by M. Nielsen and U. Engberg. Berlin, Heidelberg: Springer Berlin Heidelberg, 2002, pp. 342–356. ISBN: 978-3-540-45931-6.

[110] G. Plotkin and J. Power. "Computational Effects and Operations: An Overview". In: *Electron. Notes Theor. Comput. Sci.* 73 (Oct. 2004), pp. 149–163. ISSN: 1571-0661. DOI: 10.1016/j.entcs.2004.08.008. URL: https://doi.org/10.1016/j.entcs.2004.08.008.

[111] G. Plotkin and J. Power. "Tensors of Comodels and Models for Operational Semantics". In: *Electronic Notes in Theoretical Computer Science* 218 (2008). Proceedings of the 24th Conference on the Mathematical Foundations of Programming Semantics (MFPS XXIV), pp. 295–311. ISSN: 1571-0661. DOI: 10.1016/j.entcs.2008.10.018. URL: https://www.sciencedirect.com/science/article/pii/S157106610800412X.

[112] J. POWER and E. ROBINSON. "Premonoidal categories and notions of computation". In: *Mathematical Structures in Computer Science* 7.5 (1997), pp. 453–468. DOI: 10.1017/S0960129597002375.

[113] J. Power and E. Robinson. "Premonoidal categories and notions of computation". In: *Mathematical Structures in Computer Science* 7.5 (1997), pp. 453–468. DOI: 10.1017/S0960129597002375.

[114] V. Pratt. "Modeling concurrency with partial orders". In: *International Journal of Parallel Programming* 15 (1986), pp. 33–71.

[115] L. M. Procopio et al. "Experimental superposition of orders of quantum gates". In: *Nature Communications* 6 (2015), p. 7913.

[116] *PyWhy, An Open Source Ecosystem for Causal Machine Learning*. https://www.pywhy.org/. Accessed: 2025-10-30.

[117] T. S. Richardson and J. M. Robins. *Single world intervention graphs (SWIGs): a unification of the counterfactual and graphical approaches to causality*. Tech. rep. 128. Center for Statistics and the Social Sciences, University of Washington, 2013.

[118] E. F. Rischel and S. Weichwald. "Compositional abstraction error and a category of causal models". In: *Uncertainty in Artificial Intelligence (UAI 2021)*. 2021.

[119] P. K. Rubenstein et al. "Causal Consistency of Structural Equation Models". In: *Proceedings of the 33rd Conference on Uncertainty in Artificial Intelligence (UAI-17)*. Ed. by G. Elidan and K. Kersting. Sydney, Australia: AUAI Press, Aug. 2017. URL: http://auai.org/uai2017/proceedings/papers/11.pdf.

[120] D. B. Rubin. "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies". In: *Journal of Educational Psychology* 66.5 (1974), pp. 688–701. DOI: 10.1037/h0037350.

[121] R. Sarkis and F. Zanasi. *Graded String Diagrams for Imprecise Probability and Causal Intervention*. 2026. arXiv: 2501.18404 [math.CT]. URL: https://arxiv.org/abs/2501.18404.

[122] H. Sati and U. Schreiber. *The Quantum Monadology*. 2023. arXiv: 2310.15735 [quant-ph]. URL: https://arxiv.org/abs/2310.15735.

[123] A. Ścibior, Z. Ghahramani, and A. D. Gordon. "Practical probabilistic programming with monads". In: *Proceedings of the 2015 ACM SIGPLAN Symposium on Haskell*. 2015, pp. 165–176.

[124] A. Ścibior, O. Kammar, and Z. Ghahramani. "Functional programming for modular Bayesian inference". In: *Proceedings of the ACM on Programming Languages* 2.ICFP (2018), pp. 1–29.

[125] P. Selinger. "Towards a quantum programming language". In: *Mathematical Structures in Computer Science* 14.4 (2004), pp. 527–586. DOI: 10.1017/S0960129504004256.

[126] P. Selinger. "Dagger Compact Closed Categories and Completely Positive Maps: (Extended Abstract)". In: *Electronic Notes in Theoretical Computer Science* 170 (2007). Proceedings of the 3rd International Workshop on Quantum Programming Languages (QPL 2005), pp. 139–163. ISSN: 1571-0661. DOI: 10.1016/j.entcs.2006.12.018. URL: https://www.sciencedirect.com/science/article/pii/S1571066107000606.

[127] I. Shpitser and J. Pearl. "What counterfactuals can be tested". In: *Proceedings of the Twenty-Third Conference on Uncertainty in Artificial Intelligence*. UAI'07. Vancouver, BC, Canada: AUAI Press, 2007, pp. 352–359. ISBN: 0974903930.

[128] I. Shpitser and J. Pearl. "Complete Identification Methods for the Causal Hierarchy". In: *Journal of Machine Learning Research* 9.64 (2008), pp. 1941–1979. URL: http://jmlr.org/papers/v9/shpitser08a.html.

[129] W. Simmons. "Causal types for higher-order quantum theory". PhD thesis. University of Oxford, 2024. DOI: 10.5287/ora-qr9z52v97.

[130] W. Simmons and A. Kissinger. "Higher-order causal theories are models of BV-logic". In: *47th International Symposium on Mathematical Foundations of Computer Science (MFCS 2022)*. Vol. 241. LIPIcs. 2022, 80:1–80:14. DOI: 10.4230/LIPIcs.MFCS.2022.80.

[131] W. Simmons and A. Kissinger. "A complete logic for causal consistency". In: *arXiv preprint arXiv:2403.09297* (2024).

[132] P. Spirtes, C. Glymour, and R. Scheines. *Causation, Prediction, and Search*. The MIT Press, Jan. 2001. ISBN: 9780262284158. DOI: 10.7551/mitpress/1754.001.0001. URL: https://doi.org/10.7551/mitpress/1754.001.0001.

[133] I. Stark. "Free-Algebra Models for the $\pi$-Calculus". In: *Foundations of Software Science and Computation Structures: Proceedings of FOSSACS 2005*. Lecture Notes in Computer Science 3441. Springer-Verlag, 2005, pp. 155–169. URL: http://www.inf.ed.ac.uk/~stark/freamp.html.

[134] S. Staton. "Instances of Computational Effects: An Algebraic Perspective". In: *2013 28th Annual ACM/IEEE Symposium on Logic in Computer Science*. 2013, pp. 519–528. DOI: 10.1109/LICS.2013.58.

[135] S. Staton. "Algebraic Effects, Linearity, and Quantum Programming Languages". In: *Proceedings of the 42nd Annual ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*. POPL '15. Mumbai, India: Association for Computing Machinery, 2015, pp. 395–406. ISBN: 9781450333009. DOI: 10.1145/2676726.2676999. URL: https://doi.org/10.1145/2676726.2676999.

[136] S. Staton, H. Yang, F. Wood, C. Heunen, and O. Kammar. "Semantics for probabilistic programming: higher-order functions, continuous distributions, and soft constraints". In: *Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science*. LICS '16. New York, NY, USA: Association for Computing Machinery, 2016, pp. 525–534. ISBN: 9781450343916. DOI: 10.1145/2933575.2935313. URL: https://doi.org/10.1145/2933575.2935313.

[137] Z. Tavares, J. Koppel, X. Zhang, R. Das, and A. Solar-Lezama. "A language for counterfactual generative models". In: *International conference on machine learning*. PMLR. 2021, pp. 10173–10182.

[138] T. Verma and J. Pearl. "Equivalence and synthesis of causal models". In: *Proceedings of the Sixth Annual Conference on Uncertainty in Artificial Intelligence*. UAI '90. USA: Elsevier Science Inc., 1990, pp. 255–270. ISBN: 0444892648.

[139] T. Wang et al. *From structural causal models to higher order causal probabilistic programming: Contextual equivalences, Graded Monads and Functor Categories (Artifact)*. Oct. 2026. DOI: 10.5281/zenodo.23207221. URL: https://doi.org/10.5281/zenodo.23207221.

[140] T. Wang et al. *Typed Abstractions for Causal Probabilistic Programming*. Extended abstract presented at LAFI 2026: Workshop on Languages for Inference, co-located with POPL 2026. Non-archival workshop contribution. Rennes, France, Jan. 2026. URL: https://www.cl.cam.ac.uk/~jdy22/papers/typed-abstractions-for-causal-probabilistic-programming.pdf.

[141] T. C. Wang. "Algebraic Theories and Quantum Communication". en. MA thesis. Oxford, United Kingdom: University of Oxford, Sept. 2024. URL: https://theo.wang/works/2024-09-MSc-thesis (visited on 10/30/2024).

[142] J. Wechs, C. Branciard, and O. Oreshkov. "Existence of processes violating causal inequalities on time-delocalised subsystems". In: *Nature Communications* 14 (2023).

[143] J. Wechs, H. Dourdent, A. A. Abbott, and C. Branciard. "Quantum circuits with classical versus quantum control of causal order". In: *PRX Quantum* 2 (2021), p. 030335. DOI: 10.1103/PRXQuantum.2.030335.

[144] G. Winskel. "Event structures". In: *Petri Nets: Applications and Relationships to Other Models of Concurrency*. Vol. 255. LNCS. Springer, 1987, pp. 325–392.

[145] G. Winskel and M. Nielsen. "Models for concurrency". In: *Handbook of Logic in Computer Science*. Vol. 4. Oxford University Press, 1995, pp. 1–148.

[146] S. Witty, A. K. Lew, D. D. Jensen, and V. Mansinghka. "Bayesian causal inference via probabilistic program synthesis". In: *CoRR* abs/1910.14124 (2019). arXiv: 1910.14124. URL: http://arxiv.org/abs/1910.14124.

[147] S. A. Witty. "Bayesian Structural Causal Inference with Probabilistic Programming". PhD thesis. University of Massachusetts Amherst, 2023.

[148] M. Ying, Y. Feng, R. Duan, and Z. Ji. "An Algebra of Quantum Processes". In: *ACM Transactions on Computational Logic* 10.3 (2009), 19:1–19:36. DOI: 10.1145/1507244.1507249.

[149] A. Zhang, Q. Luo, L. Bielicke, E. Jun, and A. Chlipala. "Causality and Semantic Separation". In: *Proc. ACM Program. Lang.* 10.PLDI (June 2026). DOI: 10.1145/3808274. URL: https://doi.org/10.1145/3808274.

[150] D. Zhao and T. Fan. "Dcpo-completion of posets". In: *Theoretical Computer Science* 411.22 (2010), pp. 2167–2173. ISSN: 0304-3975. DOI: 10.1016/j.tcs.2010.02.020. URL: https://www.sciencedirect.com/science/article/pii/S0304397510001155.